

Pesticides and Adult Respiratory Outcomes in the Agricultural Health Study

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ABSTRACT: In the 1700s, Bernardino Ramazzini was among the first to describe respiratory disease among agricultural workers. Since then, farmers continue to have higher rates of respiratory illnesses, even as changes occur in occupational and environmental exposures on farms. While grain and animal exposures have been well studied for their role in agricultural lung diseases, pesticide exposures have not. Using the Agricultural Health Study, a prospective cohort study of ~89,000 licensed pesticide applicators and their spouses in Iowa and North Carolina, we are currently assessing the association of pesticides with respiratory outcomes, including wheeze, adult asthma, farmer's lung, and chronic bronchitis. At enrollment (1993–1997), 19% of farmers and 22% of commercial pesticide applicators reported wheeze in the previous year. Using logistic regression models adjusted for age, state, smoking status, and body mass index, we evaluated the association of 40 individual pesticides with wheeze within these two groups separately. In both groups, we observed strong evidence of an association of organophosphates with wheeze. For farmers, the organophosphates chlorpyrifos, malathion, and parathion were positively associated with wheeze; for the commercial applicators, the organophosphates chlorpyrifos, dichlorvos, and phorate were positively associated with wheeze. Chlorpyrifos was strongly associated with wheeze in a dose-dependent manner in both groups; use of chlorpyrifos for at least 20 days per year had an odds ratio of 1.48 (95% confidence interval [CI] = 1.00–2.19) for farmers and 1.96 (95% CI =

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1.05–3.66) for commercial applicators. Our wheeze results are consistent with recent animal models that support a role for organophosphates and respiratory outcomes.

KEYWORDS: farmers; occupational exposure; organophosphates; pesticide applicators

INTRODUCTION

Bernardino Ramazzini was among the first to identify that farmers were at higher risk of respiratory disease compared to the general population; Ramazzini's syndrome, an acute pulmonary disease of farmers associated with threshing grain, is named in his honor. Many aspects of farming have changed since the 1700s, and yet farmers continue to have higher rates of respiratory disease.¹ Pesticides may contribute to respiratory symptoms and disease. Surveys of farmers and rural residents have linked pesticides in general and particular pesticide classes with respiratory symptoms, but few have identified specific pesticides.

Evaluating the respiratory health effects of pesticides among farmers presents a challenge owing to high exposures to many other respiratory irritants and toxicants, such as animals and grains. Large sample sizes and good data on exposures to other respiratory hazards are necessary. The Agricultural Health Study (AHS), a prospective cohort study of private pesticide applicators (primarily farmers), their spouses, and commercial pesticide applicators, provides a unique opportunity to assess the respiratory hazards of pesticides. By having both farmers and commercial applicators, we can evaluate different patterns of pesticide use. To explore potential associations of pesticides with respiratory outcomes, we evaluated the common respiratory symptom wheeze and pesticide use in the past year in these two groups of pesticide applicators.

MATERIALS AND METHODS

We analyzed data from two populations participating in the AHS: private pesticide applicators (farmers) and commercial pesticide applicators. These two groups provide different perspectives on the potential impact of pesticides on respiratory health. Farmers may have exposure to other respiratory toxicants as a result of farm work, and pesticide applicators may have more days of pesticide exposure because pesticide application is their occupation, not just one aspect of their livelihood. The data from farmers were analyzed first and published in 2002,² while the data from commercial applicators was only recently analyzed to follow up on the associations observed among farmers.³ Farmers and commercial applicators completed identical questionnaires with

regard to their pesticide use and medical history. Here, we will summarize and compare the results from both populations.

The AHS is a large prospective cohort study of licensed pesticide applicators and their families in Iowa and North Carolina enrolled between 1993 and 1997.⁴ AHS participants are primarily farmers and their spouses, but almost 5000 commercial pesticide applicators in Iowa enrolled in the same manner as the farmers. Commercial applicators apply pesticides for hire to land, plants, seed, animals, waters, and structures. Details of enrollment are described elsewhere.⁴ Briefly, pesticide applicators enrolled by completing the enrollment questionnaire at the time of pesticide licensing; more than 80% of private licensed applicators and 47% of commercial pesticide applicators enrolled. Approximately 40% returned a second, more detailed questionnaire, which included items regarding respiratory outcomes. The analysis is limited to individuals who returned both questionnaires. Farmers who did and did not return the take-home questionnaire are similar with regard to demographic characteristics, farming practices, and medical history.⁵ Among the commercial applicators, however, individuals who returned the take-home questionnaire were more likely to apply and handle pesticides than those who did not; we saw no difference between the groups with regard to medical history, smoking, and demographic factors.

Exposure assessment and statistical methods were similar for both analyses, and are presented in detail elsewhere.^{2,3} Briefly, we assessed exposure and outcome using two self-administered questionnaires. These questionnaires provided information regarding 40 specific chemicals used in the year before enrollment, pesticide application methods, current agricultural activities, smoking history, medical history, and demographics. Wheeze in the past year was based on the response to the question: "How many episodes of wheezing or whistling in your chest have you had in the past 12 months?" Any positive response (one or more episodes) was defined as a wheeze case.

We focused on pesticide use in the year before enrollment because this was temporally relevant for the wheeze outcome. The questionnaire obtained information on lifetime pesticide use as well as current use. We created three level variables to evaluate lifetime use of individual chemicals: never use, use but not in the past year (former use), and current use (used in the past year). In our farmer paper,² we modeled pesticide use in a slightly different fashion; however, to ease comparison of results, we will present all results modeled as described here. For each pesticide, we evaluated both ever use in the past year and frequency of application. We used the questionnaire information on frequency of application to assess dose response.

Pesticides were evaluated using a common base logistic regression model controlling for potential confounders. Odds ratios (ORs) were not estimated for pesticides with fewer than five exposed cases. The base models included 10-year age categories, smoking history (current, past, never), body mass index (BMI, <25, ≥ 25 kg/m²), and asthma/atopy status in four levels (both, asthma

TABLE 1. Demographic, medical, and farming characteristics of farmers and commercial pesticide applicators in the Agricultural Health Study at enrollment (1993–1997)

	Farmers (<i>n</i> = 17,920) (%)	Commercial applicators (<i>n</i> = 2,255) (%)
Wheeze in past year	19	22
Age (years)		
<31	8	21
31–40	22	34
41–50	26	27
51–60	23	12
>60	21	6
Sex		
Female	3	5
Male	97	95
Race		
White	98	99
Other race	2	1
State		
Iowa	69	100
North Carolina	31	0
BMI (kg/m ²)		
<23	11	13
23–24.9	16	17
25–26.9	23	22
27–31	33	30
>31	18	18
Smoke status		
Never	55	48
Past	32	30
Current	13	23
Drink alcohol in past year	68	86
Asthma/atopy status		
Neither	86	86
Atopy only	9	9
Asthma only	3	2
Both	2	3
Current farm activities		
Own or work on farm	96	31
Raise animals	60	18
Raise crops	91	25
Pesticide use history		
Years applied pesticides		
0	1	5
<1	2	8
2–5	10	25
6–10	14	22
11–20	33	27
21–30	26	11
>30	15	4
Days applied per year		
None	1	5

Continued.

TABLE 1. Continued

	Farmers (<i>n</i> = 17,920) (%)	Commercial applicators (<i>n</i> = 2,255) (%)
<5	17	11
5–9	25	11
10–19	31	15
20–39	18	22
40–59	4	13
60–150	3	19
>150	0	5

alone, atopy alone, neither). State was included in models for farmers, but not for commercial applicators because all lived in Iowa. Previously published models for farmers did not control for BMI; models reported here include BMI. As a result of including BMI as a covariate, our sample size is reduced to 17,920 farmers compared to the 20,468 presented previously.²

Pesticides are commonly used together and can be applied to crops and animals that may independently trigger asthma symptoms. For chemicals with evidence of confounding by other pesticides, the models were run again containing the chemical of interest and the use status for the potential confounder (current, former, never). As a result of this analysis, all models for commercial applicators control for use of chlorimuron-ethyl.

The farmer analysis used the prerelease data set dated February 2000, and the commercial applicator analysis used P1REL0310 release of the AHS data set, and all statistical analysis was done using SAS (Cary, NC).

RESULTS

The association of pesticides and wheeze was evaluated among 17,920 farmers and 2,255 commercial applicators. Wheeze is a common respiratory symptom of both farmers and commercial pesticide applicators; 19% of farmers and 22% of commercial applicators reported wheezing at least once in the year before enrollment (TABLE 1). Participants ranged in age from 18 to 88, with farmers being older on average than commercial applicators. Commercial applicators were only enrolled in Iowa, while farmers were recruited from both North Carolina and Iowa. Commercial applicators were more likely to smoke than farmers, with 23% being current smokers compared to 13% of farmers. Farmers had more years of pesticide application, but fewer annual days of pesticide application, compared with the commercial applicators. Over 96% of farmers reported living or working on a farm at the time of enrollment, while only 31% of commercial applicators did.

Pesticide use by farmers and commercial applicators was similar with regard to the chemicals used (TABLE 2). However, some pesticides, particu-

larly herbicides, were more likely to be used by commercial applicators (e.g., chlorimuron-ethyl 24% versus 12%) than farmers. Farmers were slightly more likely to report some insecticides (dichlorvos, terbufos), but not all; diazinon use was more common among commercial applicators. Even among chemicals used more by farmers, commercial applicators who used these chemicals applied them more often.

Both herbicides and insecticides were associated with wheeze among farmers and commercial applicators (TABLE 3). For farmers, 5 of 40 pesticides used in the past year were significantly associated with wheeze. This is fewer chemicals than previously reported owing to the reduction in sample size by including BMI in the models; however, the OR estimates were essentially unchanged, with all but trichlorfon changing less than 2% from those reported previously. For commercial applicators, 3 of 36 pesticides (chlorimuron-ethyl, dichlorvos, and phorate) were associated with wheeze. Among commercial applicators, seven other herbicides were significantly associated with wheeze prior to adjustment for chlorimuron-ethyl use; inclusion of chlorimuron-ethyl in these models attenuated these observed OR to the null value, 1.0.³ We saw no evidence of confounding by chlorimuron-ethyl (or any other chemical) for farmers; thus, these models do not adjust for other chemical use. In general, the significant OR estimates for farmers were much lower (ORs range from 1.13 to 1.37) than commercial applicators (ORs range from 1.62 to 2.48). The pesticides with the highest ORs were organophosphates, parathion for farmers (OR = 1.37, 95% confidence interval [CI] = 0.93, 2.03) and dichlorvos (OR = 2.48, 95% CI = 1.08, 5.66), and phorate (OR = 2.35, 95% CI = 1.36, 4.06) for commercial applicators. Chlorimuron-ethyl was significantly associated with wheeze among commercial applicators but not farmers with the OR much greater among commercial applicators (OR = 1.62 versus 1.08).

We assessed dose response for chlorimuron-ethyl and two organophosphate insecticides (chlorpyrifos and phorate) with sufficient use by both farmers and commercial applicators to allow for dose-response modeling (TABLE 4). We observed significant dose-response trends for both chlorimuron-ethyl and chlorpyrifos among both groups, while phorate was only significant among commercial applicators. Use of chlorimuron-ethyl was much more frequent among commercial applicators, as a result, we were able to estimate five exposure levels (<5 to 40+ days) compared to three levels for farmers. Increased wheeze was seen at all levels of chlorimuron-ethyl use among commercial applicators, but limited to farmers using it five or more days a year. For chlorpyrifos, the increased odds of wheeze was associated with use of 20+ days per year; the commercial applicator data allow us to assess the monotonic increase at higher dose levels (40+ days/year). Commercial applicators using chlorpyrifos more than 40 days a year had an OR of 2.40 (95% CI = 1.24, 4.65). Other dose-response models can be found in our previous papers.^{2,3}

TABLE 2. Pesticide use patterns among farmers and commercial applicators in the Agricultural Health Study at enrollment (1993–1997)

	Farmers			Commercial applicators*		
	Never %	Former %	Current use** %	Never %	Former %	Current use** %
<i>Herbicides</i>						
2,4-D	20	44	36	24	33	43
Alachlor	44	47	9	55	27	19
Atrazine	26	44	30	48	26	26
Butylate	73	26	1	74	22	4
Chlorimuron-ethyl	68	20	12	62	14	24
Cyanazine	56	34	10	56	22	22
Dicamba	46	32	22	40	26	34
EPTC	79	18	3	66	21	14
Glyphosate	22	37	42	20	30	50
Imazethapyr	54	15	30	61	12	27
Metolachlor	52	29	19	54	19	26
Metribuzin	62	34	4	61	23	16
Paraquat	84	12	3	77	16	7
Pendimethalin	63	24	13	53	21	26
Petroleum oil	78	14	7	73	14	13
Trifluralin	45	37	18	51	25	24
<i>Insecticides</i>						
<i>Carbamates</i>						
Aldicarb	92	5	3	97	2	1
Carbaryl	57	31	12	60	30	11
Carbofuran	71	27	2	83	11	5
<i>Organophosphates</i>						
Chlorpyrifos	56	27	16	66	20	15
Coumaphos	90	7	3	97	3	1
Diazinon	79	15	6	73	17	10
Dichlorvos	88	9	3	92	7	1
Fonofos	77	19	4	86	10	4
Malathion	35	48	17	44	40	16
Parathion	92	7	1	96	3	1
Phorate	69	28	3	84	13	3
Terbufos	60	28	12	82	12	6
Trichlorfon	99	1	0	94	3	3
<i>Other insecticides</i>						
Lindane	86	13	1	92	7	1
Permethrin (animals)	87	9	5	92	5	3
Permethrin (crops)	86	9	5	76	12	12
<i>Fungicides</i>						
Benomyl	92	6	2	89	8	3
Captan	88	5	7	94	4	3
Chlorothalonil	92	4	3	88	5	6
Maneb	92	6	2	94	4	2
Metalaxyl	81	12	7	90	6	4
Ziram	99.5	0.3	0.2	99.7	0.2	0.1
<i>Fumigants</i>						
Aluminum phosphide	97	3	1	86	8	6
Methyl Bromide	85	11	4	96	2	1

*Results from commercial applicators previously presented in Hoppin *et al.* ³

**Current use defined as ever use in the year prior to enrollment.

TABLE 3. Chemical specific odds ratio for wheeze in the past year for farmers and commercial pesticide applicators in the Agricultural Health Study Cohort, 1993–7

Chemical	Farmers (<i>n</i> = 17,920)			Commercial applicators (<i>n</i> = 2,255)		
	Odds ratio*	95% Confidence interval		Odds ratio**	95% Confidence interval	
<i>Herbicides</i>						
2,4-D	0.97	0.86	1.10	0.99	0.73	1.34
Alachlor	1.23	1.06	1.41	0.81	0.55	1.18
Atrazine	1.18	1.05	1.32	0.91	0.63	1.31
Butylate	1.19	0.81	1.75	1.08	0.62	1.88
Chlorimuron-ethyl	1.08	0.96	1.23	1.62	1.25	2.10
Cyanazine	1.05	0.91	1.21	0.80	0.55	1.18
Dicamba	1.05	0.93	1.18	0.78	0.58	1.07
EPTC	1.37	1.08	1.73	0.90	0.61	1.32
Glyphosate	1.05	0.94	1.17	1.14	0.83	1.57
Imazethapyr	1.04	0.93	1.15	1.03	0.71	1.50
Metolachlor	1.09	0.97	1.21	1.01	0.71	1.43
Metribuzin	1.07	0.88	1.31	0.99	0.66	1.48
Paraquat	1.22	0.98	1.51	0.93	0.59	1.47
Pendimethalin	1.02	0.91	1.16	0.99	0.70	1.41
Petroleum oil	1.26	1.09	1.47	1.18	0.83	1.66
Trifluralin	1.15	1.02	1.30	0.85	0.60	1.20
<i>Insecticides</i>						
<i>Carbamates</i>						
Aldicarb	0.99	0.78	1.24	—		
Carbaryl	1.13	0.99	1.29	1.11	0.78	1.58
Carbofuran	1.14	0.88	1.48	0.77	0.46	1.30
<i>Organophosphates</i>						
Chlorpyrifos	1.09	0.97	1.23	1.27	0.92	1.74
Coumaphos	0.95	0.75	1.22	1.94	0.63	5.96
Diazinon	1.10	0.93	1.31	0.84	0.57	1.23
Dichlorvos	1.13	0.88	1.46	2.48	1.08	5.66
Fonofos	1.12	0.91	1.38	1.46	0.86	2.46
Malathion	1.13	1.00	1.27	0.95	0.69	1.31
Parathion	1.37	0.93	2.03	—		
Phorate	1.02	0.80	1.31	2.35	1.36	4.06
Terbufos	1.10	0.96	1.25	1.36	0.87	2.12
Trichlorfon	2.40	0.82	6.98	0.63	0.31	1.29
<i>Other Insecticides</i>						
Lindane	0.85	0.59	1.21	0.67	0.22	2.02
Permethrin (crops)	1.08	0.89	1.31	0.89	0.61	1.29
Permethrin (animals)	1.28	1.06	1.55	1.37	0.70	2.68
<i>Fungicides</i>						
Benomyl	1.09	0.83	1.44	1.06	0.57	1.97
Captan	1.03	0.88	1.22	0.83	0.38	1.79
Chlorothalonil	0.88	0.70	1.11	0.97	0.61	1.54
Maneb/Mancozeb	1.11	0.86	1.43	0.61	0.24	1.55
Metalaxyl	1.15	0.98	1.35	0.71	0.39	1.30
Ziram	1.06	0.48	2.36	—		
<i>Fumigants</i>						
Aluminum phosphide	0.71	0.40	1.28	1.32	0.84	2.07
Methyl bromide	1.14	0.92	1.41	—		

*Odds ratios adjusted for age, BMI, smoking, asthma/atopy and previous use of pesticide. Commercial applicators models include adjustment for chlorimuron-ethyl; farmer models include state.

Odds ratios were not estimated for pesticides with <5 exposed cases.

**Selected odds ratios for commercial applicators were presented previously.³

TABLE 4. Selected pesticide-specific dose-response models for wheeze among pesticide applicators from the Agricultural Health Study, 1993–1997.

Pesticide	Farmers (<i>n</i> = 17,920)			Commercial applicators (<i>n</i> = 2,255)		
	Odds Ratio	95% Confidence interval	<i>P</i> -trend**	Odds Ratio	95% Confidence interval	<i>P</i> -trend**
Herbicides						
Chlorimuron-ethyl			<0.01			0.01
None	1.00			1.00		
<5 days	0.95	0.80 1.12		1.87	0.96 3.63	
5–9 days	1.26	1.02 1.56		1.38	0.75 2.52	
10–19 days	1.39	1.03 1.88		1.97	1.34 2.90	
20–39 days	–			1.41	0.94 2.13	
40+ days	–			1.50	0.81 2.79	
Organophosphate insecticides						
Chlorpyrifos			0.01			<0.01
None	1.00			1.00		
<5 days	1.00	0.84 1.18		1.00	0.56 1.80	
5–9 days	1.27	1.06 1.51		1.10	0.58 2.08	
10–19 days	0.91	0.71 1.17		0.77	0.39 1.49	
20–39 days	1.48	1.00 2.19		1.96	1.05 3.66	
40+ days	–			2.40	1.24 4.65	
Phorate			0.72			0.01
None	1.00			1.00		
<5 days	1.31	0.86 2.00		2.01	0.64 6.32	
5–9 days	0.73	0.47 1.16		3.67	1.25 10.80	
10+ days	1.09	0.71 1.66		2.10	1.04 4.25	

All models adjusted for: age, BMI, smoking status, and asthma-atopy status.

Commercial applicator models include adjustment for chlorimuron-ethyl.

Farmer models are adjusted for state.

**P*-trend based on χ^2 test of trend using the categories presented.

DISCUSSION

Epidemiologic and animal data suggest that organophosphate insecticides have a role in respiratory outcomes.^{6–9} Self-reported asthma was associated with organophosphate and carbamate use among Canadian farmers, with the stronger association for carbamate insecticides.⁷ Respiratory symptoms were four times more common among Kenyan farm workers with acetylcholinesterase inhibition, a marker of organophosphate and carbamate exposure.⁸ The AHS provides the first opportunity to look at individual pesticides in detail. Among the ~20,000 AHS farmers, increased wheeze was associated with three organophosphate insecticides (parathion, chlorpyrifos, and malathion); parathion had the highest odds of wheeze among all pesticides studied.² Among the ~2000 AHS commercial applicators, increased wheeze was associated with three organophosphates (chlorpyrifos, dichlorvos, and phorate); phorate and chlorpyrifos demonstrated dose-response trends.

Pesticides have not been well characterized with regard to their respiratory toxicity. In guinea pigs, organophosphate insecticides cause airway hyperactivity.^{6,9,10} This reaction may be associated with acetylcholinesterase inhibition of the vagal nerve.^{7,10} A recent animal model suggests, however, that

airway hyperreactivity may occur at doses below those that inhibit acetylcholinesterase.^{6,9,10} This new model suggests that organophosphate-induced airway hyperreactivity results from effects of autoinhibitory M2 muscarinic receptors on the parasympathetic nerves in the lung and not from acetylcholinesterase inhibition nor from dysfunction of M3 muscarinic receptors in airway smooth muscle.⁶ If this model is appropriate for humans, it suggests that systemic exposure to organophosphate insecticides results in increased airway responsiveness at levels below those that cause acetylcholinesterase inhibition.

Chlorimuron-ethyl was strongly associated with wheeze among commercial applicators, and it confounded the associations of all other herbicides and attenuated the estimates for many of the organophosphate insecticides. There is little previous evidence of an association of chlorimuron-ethyl, or herbicides in general, with respiratory outcomes. Among farmers, we observed increased wheeze with six herbicides, including chlorimuron-ethyl; we also observed a dose-response trend for chlorimuron-ethyl. However, we saw no evidence for confounding by chlorimuron-ethyl, other pesticides, or other agricultural exposures among farmers. Chlorimuron-ethyl use was almost twice as common among commercial applicators as that among farmers and the frequency of use was greater as well. Chlorimuron-ethyl is a sulfonylurea postemergent herbicide and is only available as a dry formulation,¹¹ which may make it more likely to result in exposure via the respiratory route. Because pesticide products contain both the pesticide active ingredient as well as a large percentage of "other ingredients," we are unable to determine whether chlorimuron-ethyl itself or one of the other ingredients in the pesticide product was responsible for the association with wheeze.

We observed higher ORs among commercial applicators than farmers. This observation may be due to a number of factors including (a) more frequent pesticide use among commercial applicators, (b) lower background exposures among commercial applicators to respiratory toxicants from the farm, and (c) a healthy worker effect among the farmers. Commercial pesticide application is a transient occupation, while farming is more often a lifelong commitment; thus, individuals who farm are possibly self-selected with regard to exposures, with sensitive individuals leaving farming or altering their farming practices to minimize exposure to respiratory triggers. Our cross-sectional analysis cannot rule out this phenomenon, which has been observed in other respiratory studies of farmers.¹²⁻¹⁴ While the commercial applicators smoked more, it is unlikely that smoking influenced the higher ORs observed because commercial applicators were compared to other commercial applicators, and we made extensive efforts to control for the potential respiratory consequences of smoking. Smoking was unrelated to specific pesticide use among both farmers and commercial applicators.

The AHS provides a unique opportunity to explore associations between pesticides and respiratory outcomes. By studying both farmers and commercial

applicators, we are able to explore the role of pesticides and wheeze in populations with similar but different patterns of exposure. The associations were stronger in commercial applicators than farmers potentially because of the increased frequency of pesticide use. Our results add to the emerging literature linking organophosphate insecticides to adverse respiratory outcomes and suggest a role for chlorimuron-ethyl.

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